

DHANESH GUPTA

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PROFESSIONAL SUMMARY

Electrical Engineering undergraduate at NIT Jalandhar with interests in DevOps, computer networking, cybersecurity, backend development, and IoT systems. Building practical experience through software and hardware-oriented projects, including campus automation, network-traffic analysis, and sensor-integrated robotics. Currently strengthening Linux, networking, programming, and systems fundamentals.

EDUCATION

Dr. B. R. Ambedkar National Institute of Technology (NIT) Jalandhar

B.Tech in Electrical Engineering | Expected Graduation: 2029

Jalandhar, Punjab, India

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, C (foundational)

Web & Backend: HTML, CSS, JavaScript, Firebase, Python backend/API development (learning)

Networking & Security: Networking fundamentals, TCP/IP, GNS3, Kali Linux in lab environments, network traffic analysis

DevOps & Tools: Git, GitHub, Linux fundamentals, VS Code, command-line workflows

IoT & Hardware: ESP32, sensor integration, motor drivers, embedded systems prototyping

TECHNICAL PROJECTS

AI-Based Network Attack Forecasting (SIH 2153 project)

Python, GNS3, Kali Linux, CICFlowMeter, network traffic analysis

- Exploring a network-traffic analysis workflow to study attack patterns and build machine-learning-based detection/forecasting capabilities.
- Working with simulated network environments and traffic-derived features; evaluating the limitation of datasets without timestamps or event IDs for pre-attack forecasting.

IoT Sewer Crawler

ESP32, embedded C++, sensors, motor drivers, IoT dashboard

- Developing a concept for a sensor-equipped crawler to monitor underground environments and support remote observation.
- Working through multi-motor control, gas/air-quality sensing, distance measurement, water-level monitoring, and camera/dashboard integration.

CampusFlow — Campus Resource Automation

HTML, CSS, JavaScript, Firebase, Python (planned/in progress)

- Designing a campus-focused platform concept for classroom/lab booking, maintenance requests, event coordination, and resource workflows.
- Planning student-facing interfaces and backend/data integration to make common campus processes easier to manage.

SecureCode AI

Python, FastAPI, software security (project in progress)

- Exploring an AI-assisted code-security application and backend API structure.
- Working on application setup, dependency management, and the foundations needed for a maintainable security-focused tool.

ACHIEVEMENTS & ACTIVITIES

- 2nd Runner-Up, Forge 1.0 Buildathon (organized by the Seed Society).
- Participated in inter-college hackathon activities and collaborative project development.